



Summer Examinations 2019/2020

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This Examination, CS402/MA492, is an Open Book examination.

Course Instance Codes	3BME1, 4BME1, 1EM1, 4BCT1, 4BMS2, 3BS9, 4BS2, 3BA1, 4BFS1
Exams	3rd and 4th year (Arts and Science)
Module Codes	CS402, MA492
Module	Cryptography
External Examiner	Prof. T. Brady
Internal Examiners	Prof. G. Ellis *Dr T. Rossmann

Instructions **Answer all four questions.**
 Each question carries 25 marks.

Duration	2 hours
No. of Pages	3 pages, including this one
Discipline	Mathematics
Requirements	No special requirements
Non-programmable Calculator	Yes

Q1. (a) The ciphertext string

JWULI_KPRKZIFLYZRFTJULI_KP

was produced using a shift cipher over the 27-letter alphabet

$$A = 0, B = 1, \dots, Z = 25, _ = 26,$$

where the final character $_$ indicates a space.

By conducting a frequency analysis of a larger sample of ciphertexts, it has been found that the plaintext symbol T corresponds to the ciphertext symbol K . Determine the *first word* of the plaintext string corresponding to the ciphertext string given above. [5 marks]

- (b) Identify $\mathbb{Z}_{27} = \{0, \dots, 25, 26\}$ with $\{A, \dots, Z, _ \}$ as in part (a). The ciphertext BQXJ was obtained using a 2-dimensional Hill cipher over $\mathbb{Z}_{27}$ with key

$$\begin{bmatrix} 25 & 26 \\ 15 & 14 \end{bmatrix}.$$

Find the corresponding plaintext. [10 marks]

- (c) Explain what is meant by a *side-channel attack*. [5 marks]
- (d) An affine cipher (of dimension 1) over $\mathbb{Z}_{200}$ is used to encrypt messages. Find the number of possible keys. [5 marks]

Q2. (a) State (without proof) *Shannon's Theorem* on perfectly secure cryptosystems. [5 marks]

- (b) Consider a symmetric cryptosystem with non-empty finite sets $\mathcal{P}$, $\mathcal{C}$, and $\mathcal{K}$ of plaintexts, ciphertexts, and keys, respectively and with encryption functions $e_k: \mathcal{P} \rightarrow \mathcal{C}$ and decryption functions $d_k: \mathcal{C} \rightarrow \mathcal{P}$ indexed by keys $k \in \mathcal{K}$. Show that $|\mathcal{C}| \geq |\mathcal{P}|$. [5 marks]

- (c) i. Explain what is meant by a *linear feedback shift register (LFSR)*.
ii. Discuss whether each LFSR is determined by its connection polynomial.
iii. Consider the LFSR of length 4 with connection polynomial $1 + X + X^2$. Is this LFSR non-singular? [10 marks]

- (d) A linear feedback shift register of length 3 with connection polynomial

$$C(X) = 1 + X^2 + X^3$$

over $\mathbb{Z}_2$ is used to produce a pseudo-random sequence of binary digits with initial values $s_0 = 1$, $s_1 = 1$, and $s_2 = 0$. Determine the least period of the resulting infinite sequence $s_0, s_1, \dots$. [5 marks]

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- Q3.** (a) i. Describe the key generation, encryption, and decryption steps of the *RSA* cryptosystem.
 ii. Is $(n, e) = (78, 21)$ a valid RSA public key? [5 marks]
- (b) Alice encrypts the message $m = 140$ using the RSA cryptosystem and Bob's public key $(n, e) = (143, 7)$. Find the corresponding ciphertext (as an element of $\mathbb{Z}_{143}$). [10 marks]
- (c) Briefly explain the meaning of the following statement: "*The existence of an efficient algorithm for integer factorisation would render the RSA cryptosystem insecure.*" [5 marks]
- (d) i. Describe the *Fermat primality test*.
 ii. Can the Fermat primality test prove that 561 is composite? [5 marks]

- Q4.** (a) i. State (without proof) Gauss's characterisation of those $n \geq 1$ for which the group $\mathbb{Z}_n^\times$ is cyclic.
 ii. Which of the groups $\mathbb{Z}_{98}^\times$, $\mathbb{Z}_{99}^\times$, $\mathbb{Z}_{100}^\times$, and $\mathbb{Z}_{101}^\times$ are cyclic? [5 marks]
- (b) Alice and Bob use the cyclic group $\mathbb{Z}_{49}^\times$ and its generator $g = 3$ to perform a *Diffie-Hellman key exchange*. Alice secretly chooses $a = 5$ and Bob secretly chooses $b = 4$. Determine their shared key (as an element of $\mathbb{Z}_{25}$). [5 marks]
- (c) Describe the key generation, encryption, and decryption steps of the *ElGamal* cryptosystem. [5 marks]
- (d) Let E be the set of points of an elliptic curve over $\mathbb{Q}$ defined by the equation

$$y^2 = x^3 + ax + b.$$

Describe (either geometrically or algebraically) the usual operation of addition on E which turns E into an abelian group. [10 marks]